

Supplementary Material: Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

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This supplementary material provides additional methodological details and supporting simulation results for the primary manuscript. We describe the empirical criteria used to define the locking-region boundary, the Fourier-domain extraction procedure used to calculate modulation responses, and additional simulations showing how the locking maps depend on linewidth enhancement factor and injected-laser operating threshold. We also include time-domain AM and PM modulation traces to illustrate the extraction of AM→AM, AM→PM, PM→AM, and PM→PM response channels and to visualize nonlinear behavior at large phase-modulation depth.

I. SUPPLEMENTARY METHODS

A. Defining the Locking Range

After plotting the parameter maps of $\Delta I/\bar{I}$ and $\Delta\phi$, a fairly clear transition can be observed in both cases: one side corresponds to a low-value region with comparatively steady behavior, whereas the other corresponds to a region with increased residual oscillation or phase drift. To better identify the approximate range of this transition, we tested a series of threshold values and found that the boundary in the present simulations lies roughly near 0.1 for the intensity metric and 0.1 rad for the phase metric. Accordingly, we use $\Delta I/\bar{I} \leq 0.10$ and $\Delta\phi \leq 0.10$ rad as empirical operational thresholds to help delineate the boundary of the locking region. For the intensity metric, $\Delta I/\bar{I} \leq 0.10$ indicates residual variation below approximately 10% of the mean intensity, which in this work generally separates approximately steady trajectories from states with more pronounced oscillation. For the phase metric, $\Delta\phi \leq 0.10$ rad similarly corresponds to a small residual phase excursion over the analysis window, indicating that the field phase remains approximately stationary. These numerical values are therefore not intended as strict theoretical criteria for injection locking, but rather as empirical thresholds used to more clearly identify and outline the locking-region boundary in the present simulations. This distinction is relevant because prior studies have shown that optically injected lasers can exhibit oscillatory states with bounded phase and average-frequency entrainment near Hopf-mediated locking transitions, even when the intensity is no longer strictly steady-state^{1,2}. In that sense, the present thresholds are used to define a conservative practical operating window emphasizing jointly stable amplitude and phase behavior, rather than the broadest possible notion of frequency locking. **The cutoffs describe the visible dark–bright transition, not a rigorous stability boundary or a fraction of the color-bar range. Reducing both metric cutoffs from 0.10 to 0.01 changes the joint-region area by only 0.23% for $\alpha = 0$ and 0.96% for $\alpha = 5$ (Fig. 4). The same trajectories and finite observation window are used throughout. The archived unmodulated maps used for this cutoff comparison span -60 to 0 dB in 1 dB steps and -30 to 30 GHz in 100 MHz steps. The corresponding rate-equation implementation initializes each point with $E(0) = 0$ and $N(0) = N_{\text{th}} = N_0 + 1/(g_0\tau_s)$, with $R_p = 1.15N_{\text{th}}/\tau_n$ and $\kappa_{\text{scale}} = 1$. Fourth-order Runge–Kutta integration uses a 0.2 ps step for 90,000 steps (18 ns); the final 3,000 samples (0.6 ns) provide the intensity and unwrapped-phase metrics. The constant injected field is $E_{\text{inj}} = A_{\text{fr}}10^{R_{\text{inj,dB}}/20}$, where A_{fr} is the model free-running amplitude.**

B. Fourier-Domain Extraction of Modulation Response

For each operating point $(\Delta v, R_{\text{inj}}, f_{\text{mod}})$, the rate equations are first integrated through a transient interval, and the modulation response is extracted from the steady-state measurement window. The simulated injected-laser field is written as

$$\mathbf{E}_{\text{injected}}(t) = \text{Re}\{\mathbf{E}_{\text{injected}}(t)\} + i\text{Im}\{\mathbf{E}_{\text{injected}}(t)\}. \quad (1)$$

The amplitude response channel is defined as the normalized field-amplitude fluctuation,

$$a_s(t) = \frac{|\mathbf{E}_{\text{injected}}(t)|}{\langle |\mathbf{E}_{\text{injected}}(t)| \rangle} - 1, \quad (2)$$

where $\langle \cdot \rangle$ denotes the time average over the measurement window. The phase response channel is obtained from the optical phase,

$$\phi_s(t) = \text{unwrap} [\arg(\mathbf{E}_{\text{injected}}(t))] . \quad (3)$$

For the modulation-response extraction, the unwrapped phase traces are linearly detrended over the measurement window to remove any residual average phase drift before extracting the modulation component.

The fundamental response amplitude at the imposed modulation frequency is extracted by projecting each input and output waveform onto the sine and cosine components at f_{mod} . Specifically, for a waveform $y(t)$ sampled over the measurement window, the amplitude at f_{mod} is obtained from the quadrature components at angular frequency $2\pi f_{\text{mod}}$. This procedure is equivalent to extracting the Fourier component at the imposed modulation frequency and avoids using broadband spectral features unrelated to the driven response. The resulting input and output amplitudes are denoted as $A_{\text{in}}(f_{\text{mod}})$ and $A_{\text{out}}(f_{\text{mod}})$, respectively.

The response is reported in decibels as

$$H_{\text{dB}}(f_{\text{mod}}) = 20 \log_{10} \left[\frac{A_{\text{out}}(f_{\text{mod}})}{A_{\text{in}}(f_{\text{mod}})} \right]. \quad (4)$$

For AM-driven injection, A_{in} is extracted from the normalized remote amplitude modulation. For PM-driven injection, A_{in} is extracted from the unwrapped and linearly detrended remote phase modulation. The same procedure is applied to the corresponding AM and PM output channels to obtain the AM $\rightarrow$ AM, AM $\rightarrow$ PM, PM $\rightarrow$ AM, and PM $\rightarrow$ PM responses.

II. SUPPLEMENTARY LOCKING-REGION MAPS

The main text shows representative locking maps for the two limiting cases $\alpha = 0$ and $\alpha = 5$. To show the continuous evolution between these two limits, Fig. 1 provides the full set of two-dimensional maps for $\alpha = 0$ –5 at threshold = 1.15. The upper panel shows the relative intensity swing $\Delta I/\bar{I}$ together with the locking boundary, while the lower panel shows the corresponding binary locking-region map. These results show that the locking-region boundary evolves gradually with increasing linewidth enhancement factor, including changes in both the extent and asymmetry of the locking region.

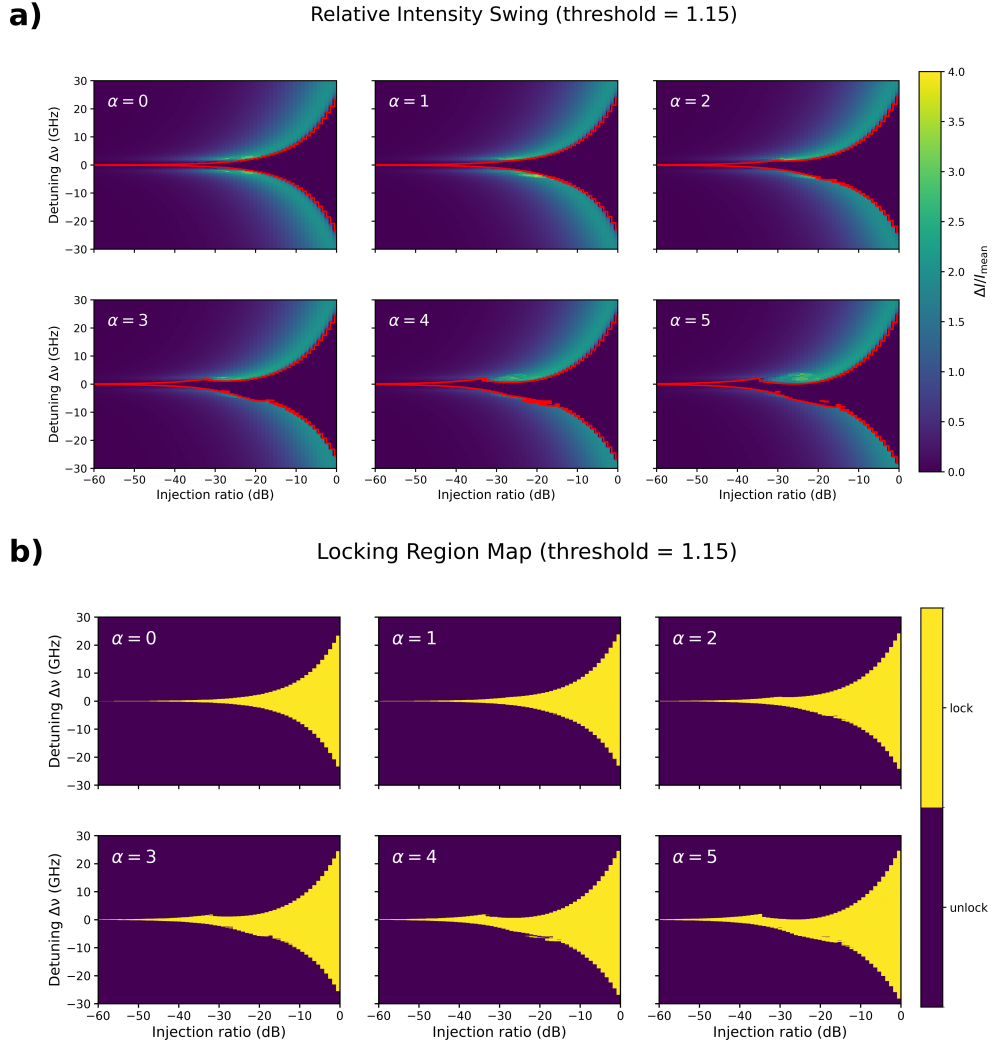


FIG. 1. Supplementary locking-region maps for threshold = 1.15. (a) Relative intensity swing maps for $\alpha = 0$ –5. The color scale shows $\Delta I/\bar{I}$, and the red contours indicate the empirical locking-region boundary defined by $\Delta I/\bar{I} \leq 0.10$ and $\Delta\phi \leq 0.10$ rad. (b) Corresponding binary locking-region maps. Yellow regions satisfy both empirical locking criteria, while purple regions do not.

Figure 2 shows the corresponding results for threshold = 1.94. Comparison with Fig. 1 shows that the locking maps depend not only on α , but also on the injected laser operating condition. In particular, changing the threshold or pump-normalized operating point modifies the locking-region extent and boundary shape as a function of injection ratio and detuning.

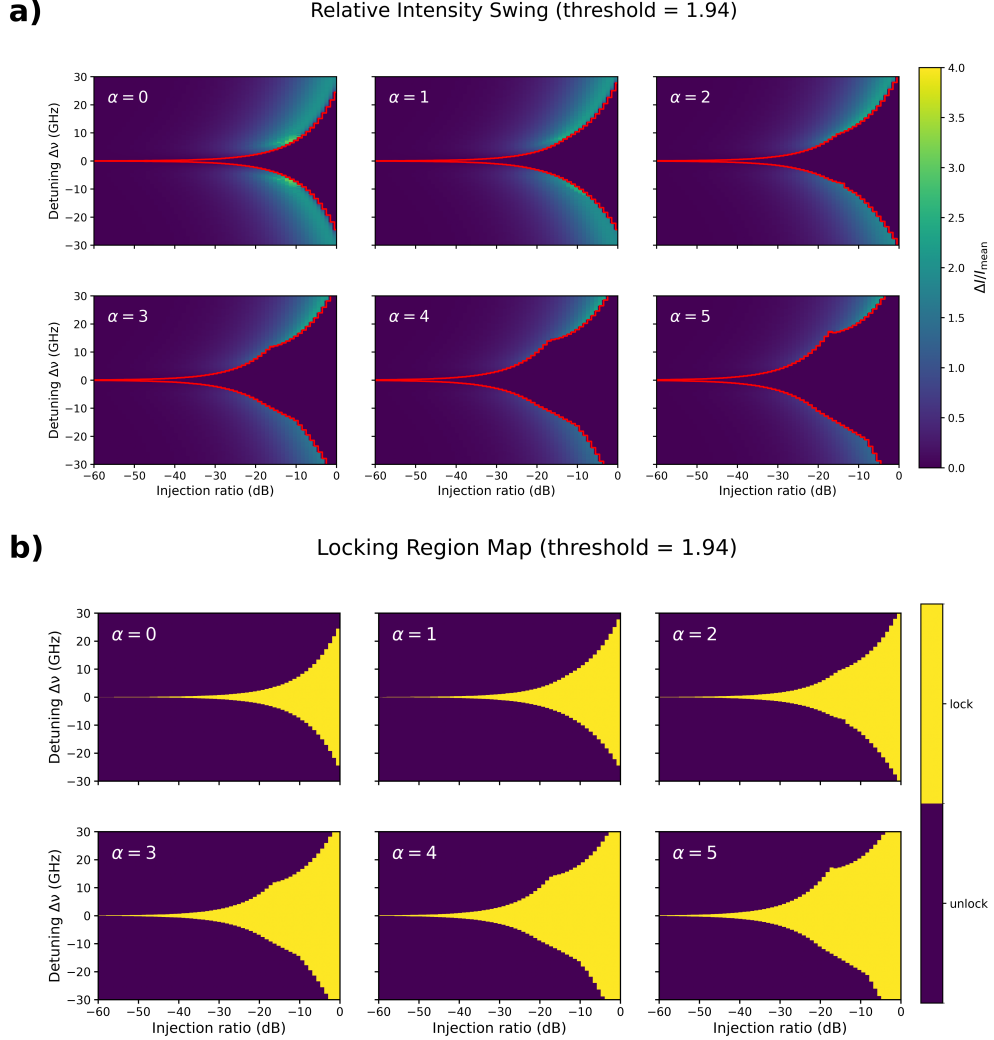


FIG. 2. Supplementary locking-region maps for threshold = 1.94. (a) Relative intensity swing maps with the empirical locking-region boundary. (b) Corresponding binary locking-region maps. Compared with the threshold = 1.15 case, the locking-region boundary and accessible locking range change with the injected laser operating condition.

III. TIME-DOMAIN ILLUSTRATION OF MODULATION CHANNELS

Figure 3 illustrates the time-domain input and output channels used for extracting the modulation response. The traces are calculated at a representative operating point with $\alpha = 5$, $R_{\text{inj}} =$

−50 dB, zero detuning, and $f_{\text{mod}} = 2$ GHz. After a transient interval, the last eight modulation cycles are plotted for modulation depths from 0.60 to 1.00. For AM-driven injection, the input channel is the normalized source-field amplitude modulation, and the output channels are the normalized injected-laser field amplitude and the injected-laser field phase. These channels correspond to the AM→AM and AM→PM responses. For PM-driven injection, the input channel is the source-field phase modulation with depth $m_\phi = 0.6\pi$ to 1.0π , and the output channels are again the injected-laser field amplitude and phase, corresponding to the PM→AM and PM→PM responses. Thus, the figure provides a direct time-domain visualization of the channels used in the Fourier-domain extraction procedure described above.

This figure also provides additional detail for the nonlinear phase-modulation behavior discussed in the main text. In the small-signal regime, increasing modulation depth would primarily scale the response amplitude. However, for PM modulation depths of approximately 0.6π and above, the PM→AM and PM→PM traces show changes in waveform shape and relative amplitude that are not simple proportional rescalings. These time-domain traces therefore help visualize the onset of nonlinear response at large PM depth. **The dimensionless PM depth is $m_\phi = \Delta\phi/\pi$; thus $m_\phi = 0.6$ – 1.0 corresponds to a phase excursion amplitude $\Delta\phi = 0.6\pi$ – 1.0π . Waveform distortion alone does not identify a bifurcation.**

AM/PM modulation responses ($\alpha = 5.0$, $R_{\text{inj}} = -50$ dB, $f_{\text{mod}} = 2$ GHz)

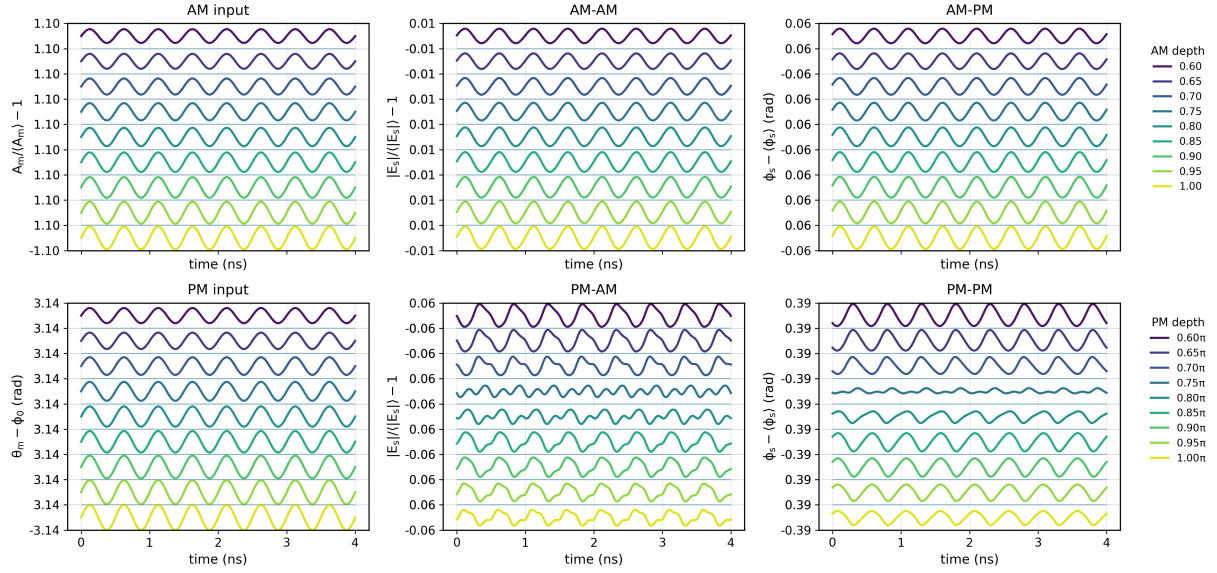


FIG. 3. Time-domain AM and PM modulation responses. The first row shows AM input, AM→AM, and AM→PM channels. The second row shows PM input, PM→AM, and PM→PM channels. The traces illustrate how the four modulation response channels are defined for Fourier-domain extraction and show the nonlinear evolution of the PM-driven response at large phase-modulation depth.

IV. ADDITIONAL NUMERICAL AND EXPERIMENTAL VALIDATION

A. Sensitivity of the Descriptive Boundary

Figure 4 overlays the joint cutoff contours on the archived intensity maps. The contours remain close when both metric thresholds are varied between 0.01 and 0.10, although individual intensity-only contours need not show the same insensitivity. The comparison uses the same stored trajectories and observation window; it is not a bifurcation or observation-time convergence test.

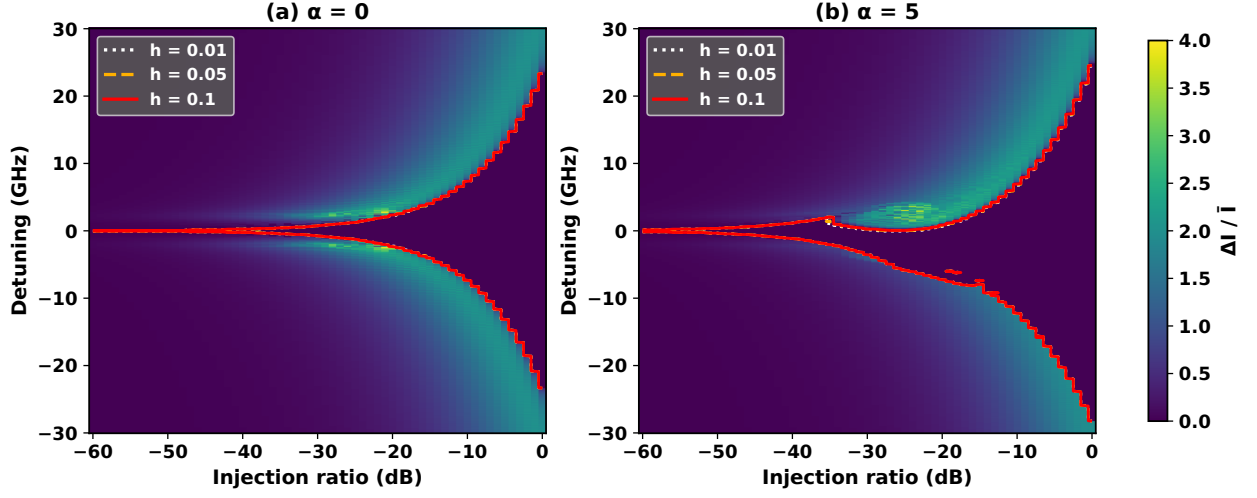


FIG. 4. **Empirical cutoff comparison.** Intensity-swing maps for a) $\alpha = 0$ and b) $\alpha = 5$, with contours of the joint conditions $\Delta I / \bar{I} \leq h$ and $\Delta \phi \leq h$ rad at $h = 0.01, 0.05$, and 0.10 . Color represents the original relative intensity swing. The comparison assesses the descriptive contour, rather than proving dynamical stability.

B. Modulation-Depth Coverage

The Fig. 5 data cover modulation depths from 1% to 100% in 1% increments. Figure 5 explicitly includes the 1%, 5%, and 10% traces alongside the representative larger depths displayed in the main text. These curves are extracted from the original stored frequency-response arrays.

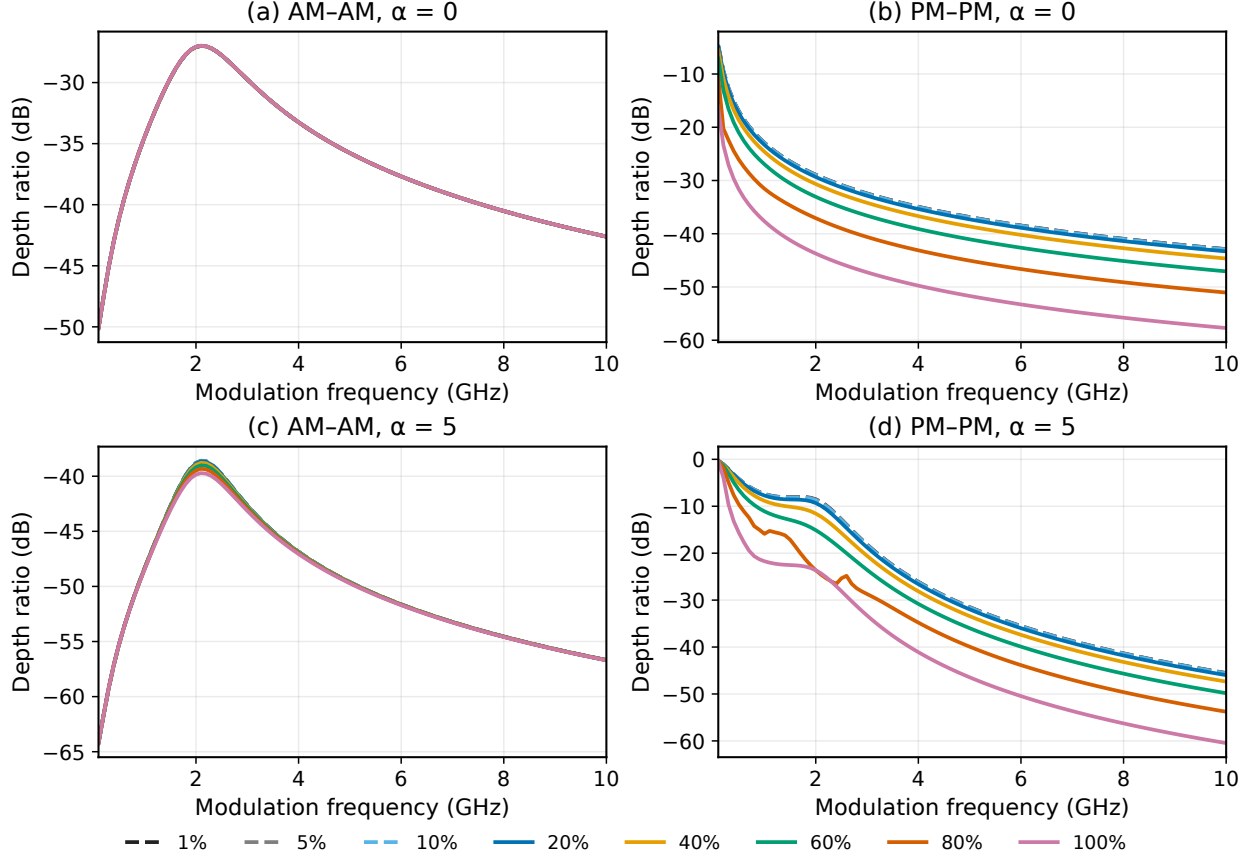


FIG. 5. **Expanded depth selection from the Fig. 5 sweeps.** AM→AM and PM→PM fundamental responses at $R_{\text{inj}} = -50$ dB and zero detuning for **a,b)** $\alpha = 0$ and **c,d)** $\alpha = 5$. The displayed depths are 1%, 5%, 10%, 20%, 40%, 60%, 80%, and 100%; the underlying grid contains every 1% increment.

C. Comparison with Fiber Measurements

Figure 6 compares fiber-based measurements under unmodulated and intensity-modulated injection with the rate-equation model. We extract the injected-laser intensity FFT for the unmodulated case and the fundamental amplitude response for the modulated case. The measurements show corresponding locking-related spectral features and a low-transfer region under modulated injection.

The AM generated by the MZM in our experiment is accompanied by PM. We therefore include an additional phase modulation in the injected field:

$$E_{\text{inj}}(t) = A_{\text{inj},0}[1 + 0.1 \sin(2\pi f_{\text{mod}}t)] \exp[i0.10 \sin(2\pi f_{\text{mod}}t)]. \quad (5)$$

The AM depth is a field-amplitude fraction; the added peak PM is 0.10 rad, in phase with the AM drive. This effective model parameter is not an independent measurement of the EOM chirp. The simulated minimum lies near -0.16 GHz for a 4 GHz drive, reproducing the negative-detuning low-response region seen experimentally.

Simulations use $\alpha = 2$ and normalized pump setting 1.944, representing injected-laser output near 0 dBm. The external experimental ratio of -15 dB and effective simulated ratio of -40 dB imply a model-dependent power-coupling estimate of approximately -25 dB; this offset is not an independent measurement of κ .

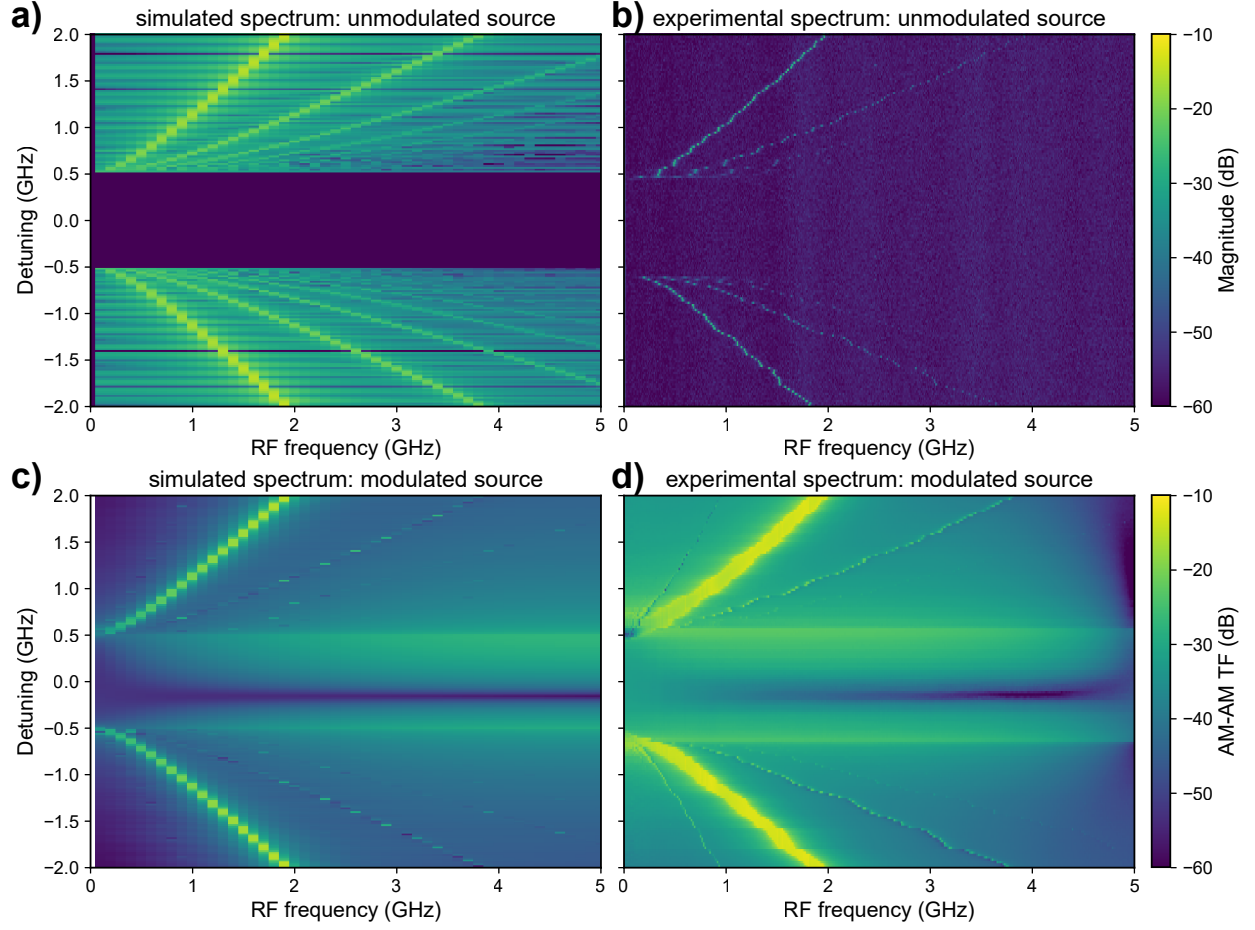


FIG. 6. Fiber-based experiment and simulation comparison. Unmodulated injection: **a)** simulated injected-laser intensity RF spectrum and **b)** measured spectrum in SVA spectrum mode. Modulated injection: **c)** simulated amplitude response including the additional 0.10 rad input PM and **d)** measured response in VNA mode after reference subtraction and optical-power normalization. The simulated fundamental fractional field-amplitude response is normalized to the 0.1 input AM depth. Horizontal axes show RF frequency; vertical axes show optical detuning. Both rows retain the original -60 to -10 dB color limits, using their respective spectral and transfer-function normalizations.

D. Complex-Field Readout and Facet Reflection

Supplementary Fig. 7 uses the same inputs as main-text Fig. 6: independent generators with seeds 4 and 8 for the remote and local paths, a 20 ns CW settling interval, and 200 active encoded symbols at 10 Gbaud. Each antipodal pair encodes one independent data magnitude, corresponding to a 5 Gbaud data-symbol rate. The zero-order-hold drives are filtered by the 20 GHz Gaussian FIR filter and sampled at 64 points per encoded symbol. Both figures display the first 100 active symbols (20–30 ns).

The local remodulator acts on the complete complex rate-equation output, including residual modulation. With an additive coherent reflected field, the model is

$$\begin{aligned} E_{\text{out}}(t) &= E_{\text{injected}}(t) + \sqrt{R_f} E_{\text{inj}}(t), \\ E_{\text{injected}}^{(\text{modulated})}(t) &= \frac{E_{\text{out}}(t)}{\langle E_{\text{injected}} \rangle_{\text{CW}}} y(t). \end{aligned} \quad (6)$$

Here $R_f = 0$ or 0.30 is a power reflectivity; the reflected field coefficient is therefore $\sqrt{R_f}$. The injection is $E_{\text{inj}} = A_{\text{fr}} 10^{R_{\text{inj,dB}}/20} [1 + x(t)]$, while the computational data arm is the signed field $E_{\text{remote}}^{(\text{modulated})} = x(t)$. The normalization is the complex mean of the no-reflection field over 10–19 ns, before modulation, and is shared by both reflection cases without fitting. Fields use the same effective output-plane units with unit intracavity-to-output conversion. This additive comparison keeps the laser parameters fixed; device-specific facet coupling and output conversion would require calibration.

Balanced detection and sample-resolved accumulation are

$$D_j = 2 \operatorname{Re} \left\{ x_j \left[E_{\text{injected},j}^{(\text{modulated})} \right]^* \right\}, \quad A_n = \frac{1}{64} \sum_{j=1}^n D_j, \quad (7)$$

starting at 20 ns. At each symbol end, this is the cumulative sum of symbol-averaged products. The ideal coherent reference uses the same filtered inputs and $D_j^{\text{ideal}} = 2x_j y_j$. No samples are omitted in the field, differential-signal, or accumulation traces. The mapping $(1 + \operatorname{Re} E)/2$ in main-text Fig. 6b is for display only and is not used for multiplication. Reflection may increase an error or partly cancel it in an individual trajectory; such cancellation does not establish improved performance.

Supplementary Fig. 7: matched main-text sequence with 30% output reflection

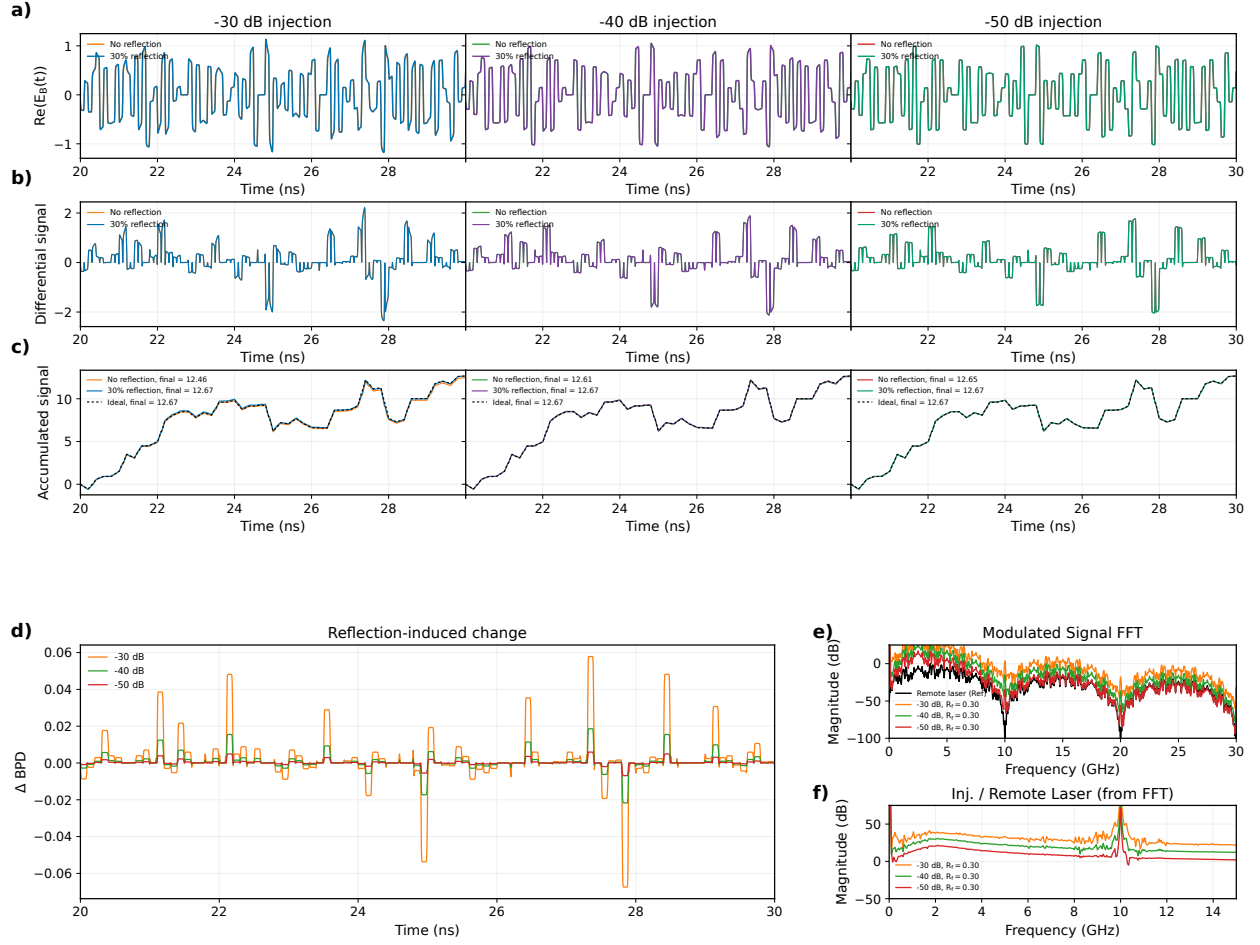


FIG. 7. Output reflection under the main-text Fig. 6 conditions. All panels use $\alpha = 0$. **a–c)** Locally remodulated fields, balanced differential signals, and accumulation over 20–30 ns, respectively, at –30, –40, and –50 dB injection. Contrasting solid curves denote $R_f = 0$ and 0.30; dotted curves in (c) are ideal coherent results. Legends in (c) give the values at 30 ns. **d)** Reflection-induced differential-signal change. **e)** Pre-remodulation spectra of the complete reflected output (carrier, residual modulation, and reflected input), with the remote spectrum as reference. **f)** Injected-to-remote spectral ratios from (e). All time-domain traces retain 64 samples per symbol.

E. Accumulated-Error Statistics over 1000 Inputs

We extend the calculation to $Q = 1000$ independent remote/local input pairs, each containing 1024 active encoded symbols. All conditions reuse the same pairs, with distinct remote and local vectors: $\alpha = 0, 2, 5$, injection ratios $-50, -40$, and -30 dB, and $R_f = 0$ or 0.30 , at zero detuning. The 100% field-modulation depth, eight-level magnitudes with antipodal pairing, filtering, sampling, and 20 ns CW preamble are unchanged. Each length L is a prefix of the same trajectory, without restarting the laser or drawing new inputs.

For input pair q , the error and ensemble RMSE after L encoded symbols are

$$\begin{aligned}\varepsilon_q(L) &= A_{\text{OIL},q}(L) - A_{\text{ideal},q}(L), \\ A_{\text{OIL},q}(L) &= \frac{1}{64} \sum_{j=1}^{64L} D_{q,j}, \quad \text{RMSE}(L) = \sqrt{\frac{1}{Q} \sum_q \varepsilon_q(L)^2}.\end{aligned}\tag{8}$$

The ideal reference uses identical filtered waveforms and perfectly coherent clean carriers. These metrics quantify additional OIL/reflection error, excluding filtering error relative to the unfiltered digital dot product.

Both panels of Fig. 8 use the same final-error percentage:

$$S = \sqrt{\frac{1}{Q} \sum_q A_{\text{ideal},q}(1024)^2} = 27.7159, \quad p_q = 100 \frac{\varepsilon_q(1024)}{S}.\tag{9}$$

Thus S is the RMS scale of all 1000 ideal endpoints, not an error. Each signed error is divided by this common scale and multiplied by 100. The same denominator is used for all parameter/reflection conditions, preserving distribution shape and avoiding division by individual near-zero ideal results. A value of $p_q = 1\%$ means an error of $0.01S$, not 1% of that trial's ideal result. Gaussian fits describe the histograms but do not establish Gaussianity.

Figure 9 also reports

$$\begin{aligned}\text{NRMSE}(L) &= \sqrt{\frac{\sum_q \varepsilon_q(L)^2}{\sum_q A_{\text{ideal},q}(L)^2}}, \\ r_P(L) &= \text{corr}_q[A_{\text{OIL},q}(L), A_{\text{ideal},q}(L)].\end{aligned}\tag{10}$$

NRMSE is displayed as a percentage. Pearson correlation measures covariation across input pairs and must be interpreted together with NRMSE: high correlation alone does not imply small numerical error.

The dependence on α reflects coupled amplitude/phase dynamics and coherent readout. Writing the calibrated no-reflection field as $E_{\text{injected}}/\langle E_{\text{injected}} \rangle_{\text{CW}} = (1 + a_\alpha)e^{i\phi_\alpha}$ gives

$$\delta D_\alpha = 2xy[(1 + a_\alpha)\cos\phi_\alpha - 1] \simeq 2xy\left(a_\alpha - \frac{\phi_\alpha^2}{2}\right). \quad (11)$$

Here a_α and ϕ_α are residual field-amplitude and phase fluctuations after CW calibration. Small AM affects the aligned readout to first order, whereas small PM enters to second order. Nonzero α couples amplitude and phase,³ but does not guarantee lower error. In the sampled conditions, RMSE decreases for nonzero α at -50 and -40 dB, while this trend reverses at -30 dB, especially for $\alpha = 5$. The simulations retain the full nonlinear rate equations and exact readout; the small-residual expansion only explains sensitivity.

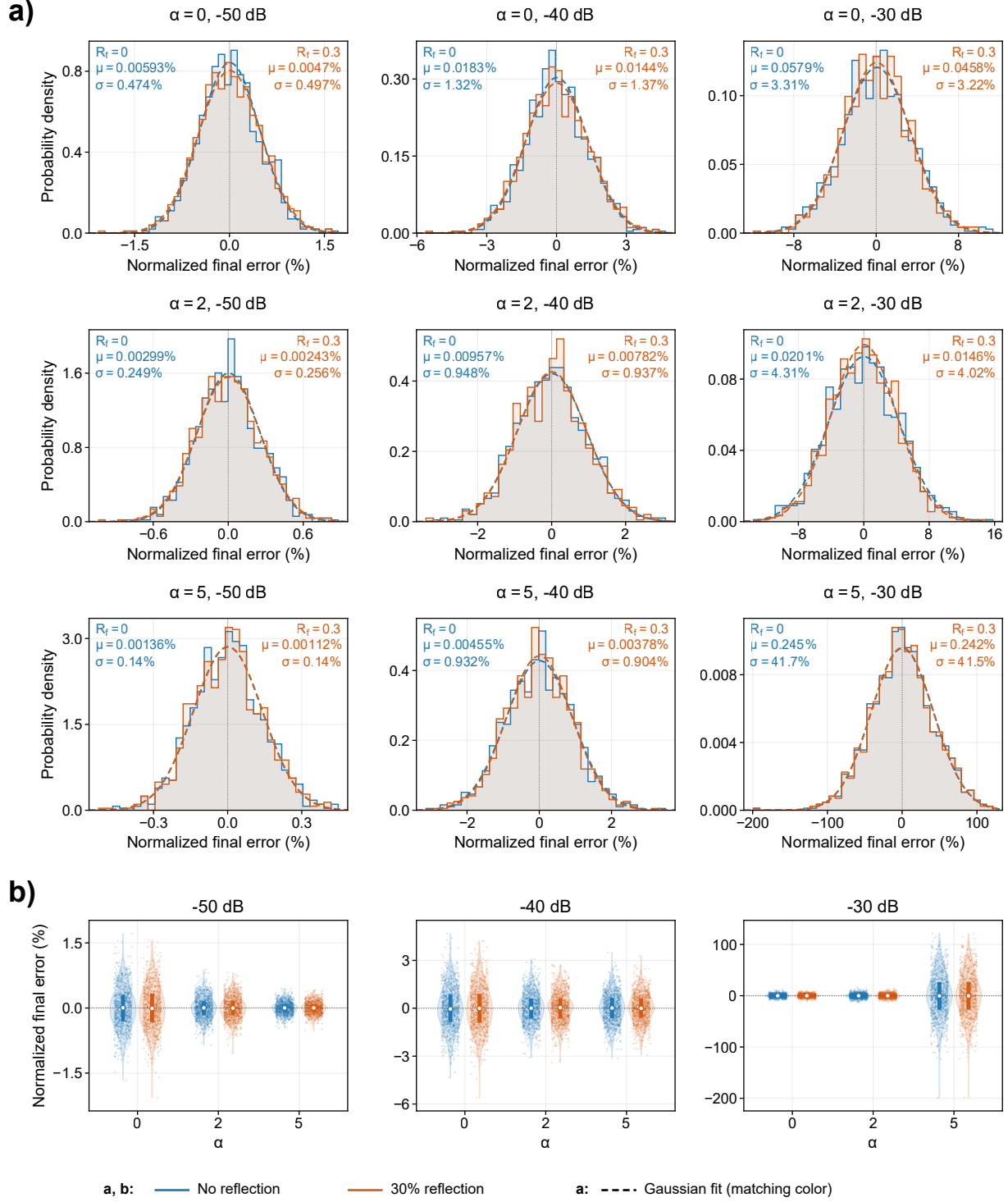


FIG. 8. Final accumulated-error distributions for 1000 inputs at $L = 1024$. **a)** Histograms of p_q (%) with descriptive Gaussian fits; fitted μ, σ are in percentage points. Rows: $\alpha = 0, 2, 5$. **b)** The same normalized errors versus α : dots denote individual inputs, violins density, thick bars the 25th-75th percentiles, and white dots medians. Columns: $-50, -40, -30$ dB. Blue/orange: $R_f = 0/0.30$; matching dashed curves in (a): Gaussian fits. Both panels use the common ideal RMS $S = 27.7159$.

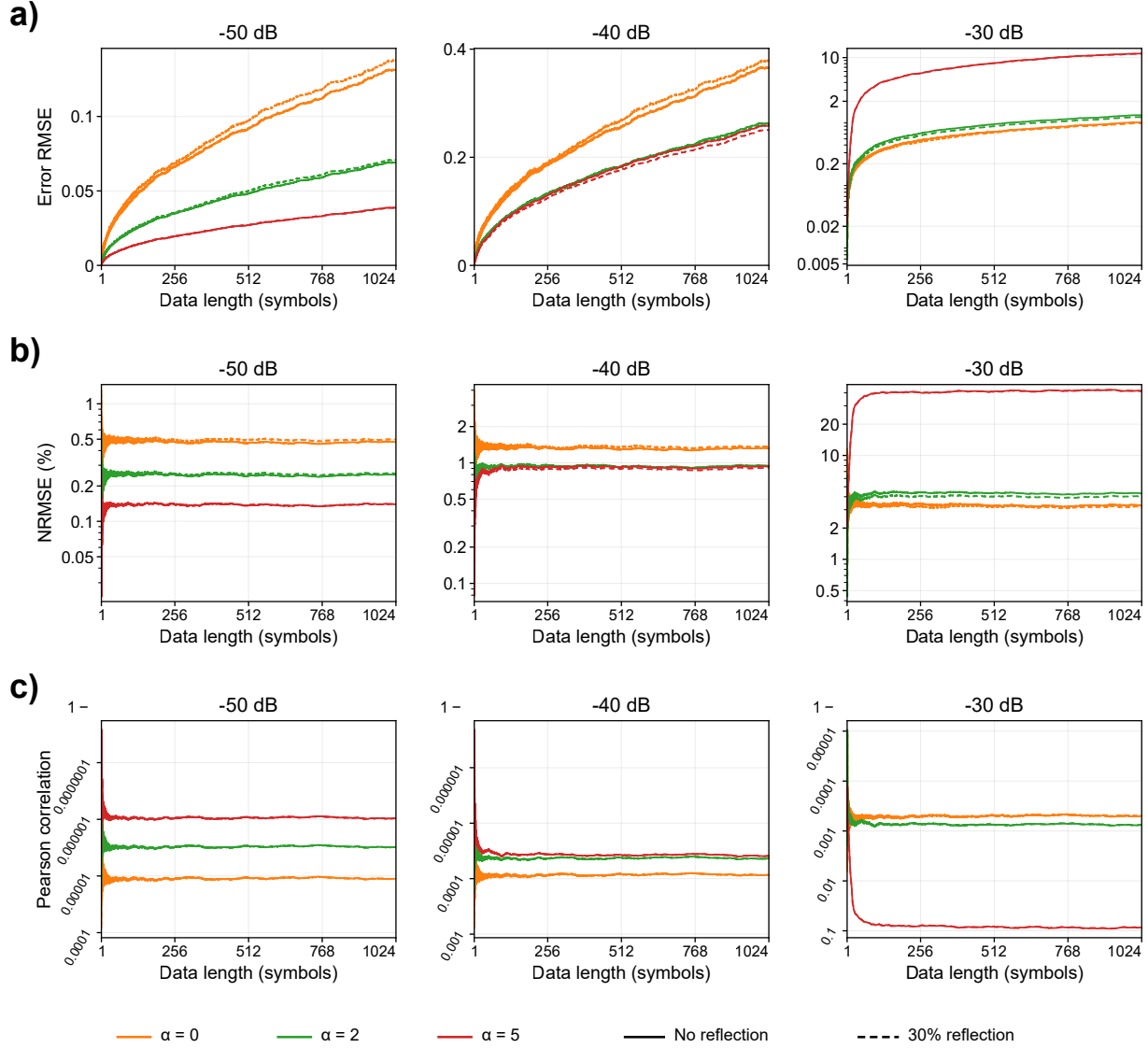


FIG. 9. **Accumulation accuracy versus prefix length for 1000 input pairs.** a) RMSE, b) NRMSE (%), and c) Pearson correlation for $L = 1$ –1024, using the inputs of Fig. 8. Columns: -50 , -40 , -30 dB; orange/green/red: $\alpha = 0/2/5$; solid/dashed: $R_f = 0/0.30$. Subplots have independent vertical ranges. RMSE axes are linear at -50 and -40 dB and logarithmic at -30 dB; NRMSE axes are logarithmic. In (c), logarithmically spaced ticks show $1 - r_p$ under a $1 -$ prefix. The reference uses identical filtered inputs and ideal coherent carriers.

V. SHORT-REACH IMPLEMENTATION AND STABILIZATION

The intended setting is short-reach interconnection within a data center, including chip-to-chip links, using established telecommunications components. This scope reduces the propagation distance and associated impairments but does not assume a noise-free channel. The measures below are a proposed implementation strategy, not a demonstrated closed-loop network.

To compensate slow relative-frequency drift, a tapped carrier/lock monitor with a frequency-selective discriminator and filtering would feed a PID controller acting on the injected-laser current and temperature. Active injection-lock stabilization by optical discrimination and current feedback has been demonstrated⁴. The reference is the carrier retained in the injected AM field; data-bearing sidebands need not provide a separate locking reference. A separate phase actuator after local remodulation and before interference would use a filtered pilot- or calibration-derived error signal to stabilize the readout phase. Its loop bandwidth would be below the symbol rate but above the drift frequencies to be corrected. This phase servo cannot substitute for maintaining the laser inside its locking range, and filtering the unknown data alone does not guarantee a usable phase-error signal.

Polarization-maintaining fiber with aligned launch axes, or active polarization tracking, would preserve coupling and polarization overlap. These measures do not eliminate acoustic or thermal optical-path phase fluctuations; compact routing, environmental isolation, and the phase servo address those disturbances within the available control bandwidth. Chromatic dispersion can be reduced by limiting link length or using compatible 1310 nm lasers and components near the zero-dispersion region of conventional single-mode fiber⁵; it is not generally zero, and changing wavelength requires reassessing the laser and link parameters. Short links can avoid in-line amplification when the received-power budget permits, thereby avoiding that source of amplifier noise. Residual laser noise, fluctuations outside the servo bandwidth, and propagation loss still require link-specific characterization. The present deterministic simulations therefore do not establish a maximum transmission distance or complete network-noise tolerance.

REFERENCES

- ¹B. Kelleher, D. Goulding, B. Baselga Pascual, S. P. Hegarty, and G. Huyet, *Physical Review E* **85**, 046212 (2012).
- ²J. Pausch, C. Otto, E. Tylaite, N. Majer, E. Schöll, and K. Lüdge, *New Journal of Physics* **14**, 053018 (2012).
- ³E. K. Lau, L. J. Wong, and M. C. Wu, *IEEE Journal of Selected Topics in Quantum Electronics* **15**, 618 (2009).
- ⁴B. Saxberg, B. Plotkin-Swing, and S. Gupta, *Review of Scientific Instruments* **87**, 063109 (2016).
- ⁵Corning Incorporated, “SMF-28 Ultra Optical Fiber: Product information,” <https://www.corning.com/media/worldwide/coc/documents/Fiber/product-information-sheets/PI-1424-AEN.pdf> (2025), pI-1424-AEN; zero-dispersion wavelength 1304–1324 nm.